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Solid State Communications

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First principles study of transport properties of $\text{LaAlO}_3/\text{SrTiO}_3$ heterostructure with water adsorbates

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ARTICLE INFO

Article history:

Received 3 May 2013

Received in revised form

11 June 2013

Accepted 9 July 2013

by P. Sheng

Available online 15 July 2013

Keywords:

A. $\text{LaAlO}_3/\text{SrTiO}_3$

C. Interface

D. Polar molecule sensor

D. Two-dimensional electron gas

ABSTRACT

Tuning transport properties of two-dimensional electron gas (2DEG) at the $\text{LaAlO}_3/\text{SrTiO}_3$ interface by surface modification is of great interest due to its rich physics and potential application in polar liquid or gas sensors. From first-principle calculations, we found that the adsorption of a thin layer of polar water molecules on the surface of LaAlO_3 can remarkably enhance the carrier density of the interfacial 2DEG by over at least 50% which is qualitatively consistent with reported experimental result. This result sheds light on the charge coupling between surface polar molecules and interfacial 2DEG and suggests that sensor devices may be built using the hetero-oxide interfacial 2DEG.

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1. Introduction

The high-mobility two-dimensional electron gas (2DEG) generated at the interface between two band insulators LaAlO_3 (LAO) and SrTiO_3 (STO) reveals intriguing electronic properties [1–8] as well as superconducting phenomenon at low temperature [9,10]. The mechanism behind this interface conduction was explained by the intrinsic polar-catastrophe model [2,3], where a charge transfer from the LAO layer to the interface is associated with a build-up of the internal electric field in the LAO layer which originates from the polar discontinuity between LAO and STO.

Recent experimental reports clearly revealed the importance of the correlation between the interface and the surface charge states, and therefore the manipulation of interfacial conduction through introduction of surface charges and adsorbates [11–15], or a capping layer such as STO [16]. The fact that the interface conducting states can be dramatically sensitive to the surface adsorbates leads to a strong possibility of LAO/STO interfaces functioning as polar molecule sensors and a new generation of electronics [11,12]. Xie et al. [17] have reported a three times change of conductivity at the LAO/STO interface induced by the surface adsorption of water. This conductivity change has been attributed to an increase in the sheet carrier density of the 2DEG by more than $2 \times 10^{13} \text{ cm}^{-2}$, suggesting that the adsorbent has a large influence on the charge transfer from the film surface to

the interface. Very recently, we reported an in-plane field effect transistor structure based on LAO/STO 2DEG with polar molecules such as water as the gate to affect source/drain current [18], and a model based on the formation of a Schottky junction between the 2DEG and the water adsorbates affected interface was proposed to interpret the rectified I - V characteristics. These observations demonstrate that such a fascinating interface system can be used for high-performance polar molecule sensors. In this letter, we report our first-principle calculations of the influence of water molecules on the electronic transport properties of $\text{LaAlO}_3/\text{SrTiO}_3$ heterostructure in order to provide a theoretical explanation to these experimental results.

2. Calculation details

First-principle calculations are performed within the framework of density functional theory and local density approximation (DFT–LDA) using the Vienna *ab initio* Simulation Package (VASP) [19,20]. The supercell shown in Fig. 1 contains 4 unit-cells of LAO on 4 unit-cells of STO substrate with the in-plane lattice constants fixed at their theoretical values, $a=b=3.8633 \text{ Å}$. For the structures with water, one (or two) water molecule is added on the top of the LAO layer. The thickness of the vacuum region between the slabs is about 20 Å. The cut-off energy of the plane wave basis is 500 eV; the k -point grid size is $11 \times 11 \times 1$; and Gaussian smearing is applied with a small smearing width 0.05 eV during calculation of energy. A dipole correction is applied along the out-of-plane axis

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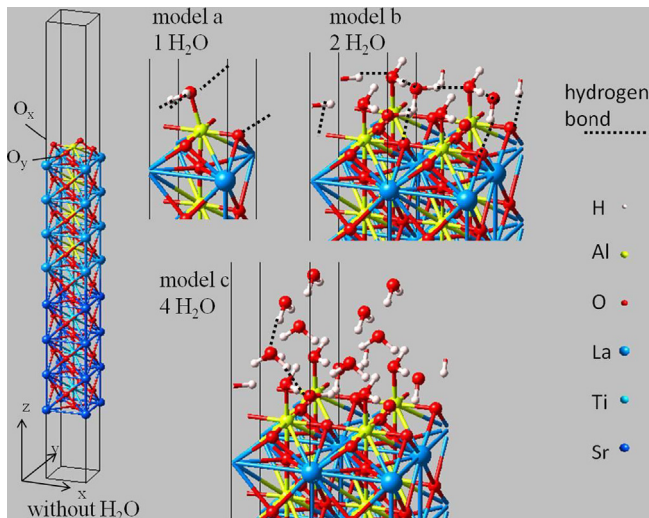


Fig. 1. (Color online) Optimized atomic structure of four models. Model a contains one molecule on (1×1) LAO surface, model b contains two molecules on (1×1) LAO surface and model c contains four molecules. Neighboring periods for models b and c are shown for convenience. Dashed lines represent hydrogen bonds, and dashed circles represent atoms shift outside the region for display. O_x and O_y represent two oxygen atoms on surface which form bonds with Al in x and y directions.

to cancel the dipole–dipole interaction due to the dipole moment in each slab [21,22].

The first consideration in building the model of H_2O on LAO is to allow the H_2O be fully optimized by the lowest energy criterion without restriction of the symmetry. Many different models have been tried, and eventually models a and b come out to have the lowest energy among all calculated models with one and two H_2O per (1×1) surface unit, respectively. Dissociative adsorption [23,24] was considered but found to be energetically unfavorable, and it should also be pointed out that the H_2O molecules were not decomposed on the surface of LAO, and the dissociation of hydrogen [25] did not occur during optimization. Having determined the first layer of water molecules bonding on the LAO with lowest energy, we put the second layer of H_2O by forming hydrogen bonds. Due to the relatively weak hydrogen bond and easy rotation of the bond (the characteristics of liquid water), it is assumed that the bonding angle will be more determined by aligning the polarization direction to the electric field direction of the LAO layer that leads to a picture suggested by Xie et al. [17].

3. Results and discussion

Fig. 1 shows optimized atomic structure of four models, where model a contains one H_2O molecule, model b contains two H_2O molecules on the (1×1) LAO surface and model c contains two layers of H_2O molecules. In the optimized structures, O_x (notation of O_x and O_y representing two O atoms on AlO_2 terminated surface shown in Fig. 1) in models a and b shift upward due to the attraction from H of H_2O . It should be noted that the structure of model a is consistent with that from other theoretical studies of H_2O molecular adsorption on the surface of other perovskites [23,24]. The initial structure of model b is similar to that of a monolayer of H_2O adsorbed on $SrTiO_3$ from Refs. [23,24], where one H_2O is attached to a B-site (Ti) atom and forms hydrogen bond with another H_2O which also forms hydrogen bond with an O ion on the LAO surface.

The binding energy (E_{binding}) per H_2O on the LAO/STO heterostructure is $E_{\text{binding}} = (E_{\text{without } H_2O} + n E_{\text{isolated } H_2O} - E_{\text{with } H_2O})/n$ and n is the number of H_2O per (1×1) unit cell. The binding energy in model a is 1.43 eV, while it is 1.40 eV in model b with two H_2O

resulting from both the adsorption of the H_2O on the surface and the hydrogen bonds between the H_2O molecules. Since the H_2O bonded with Al atom is similar to that in model a, and another H_2O is linked by hydrogen bond, the slight decrease of the binding energy indicates that the adsorption energy of H_2O should be in the same order as the hydrogen bond energy. It is also well known that LDA overestimates the binding energy in general [26,27]; therefore the actual binding energy of the adsorbed water molecule on the LAO surface should be lower, and it is expected that the water molecule can be easily desorbed by thermal energy. This explains the experimental fact that by heating the sample to over 100°C , the water molecule induced I – V characteristics [18] disappear and a normal metallic conduction of the interface is resumed.

It should be noted that models a and b have the lowest energy for one and two molecules per (1×1) unit cell. Model a is the most possible model for one monolayer of water, while model b is the most possible model for two layers of water. More layers of water molecules need a very large number of atoms in the model and a large number of calculations which are beyond our computing limit. Instead, we argue that the aligning of the water molecule along the polarization of the LAO film through hydrogen bonding with the oxygen of LAO could happen; we also include model c where four layers of water molecules align in the same direction forming a hydrogen bonding with LAO. Even though a lowest energy state is not obtained, this model can illustrate the importance of polarization alignment of water molecules.

Fig. 2(a) shows the band structure before and after adding H_2O (models a–c). The band structure in the absence of water is in agreement with those of other calculations [16,28,29] where the Ti energy bands (conduction band) at the surface layer cross the Fermi-level (E_F) around the Γ -point (0,0), and the O 2p band (valence band) crosses E_F at M. These indicate the presence of conducting electrons in STO at the LAO/STO interface and holes are left at the surface of LAO. The hole states however have not been reported to contribute to the electrical transport on the surface of LAO/STO. The holes may be localized as a result of surface impurities, surface roughness or corrected band theory [28]. It can be seen that there is no significant change to the conduction band structure after adding H_2O . However, the hole states, in the valence band at M (0.5,0.5) shift downward in models a and b. The valence band of models a and b rises at X (0,0.5) or X' (0.5,0) as shown in Fig. 2, and this change may depend on the orientation of H_2O molecule on the top surface where the H atoms of H_2O are

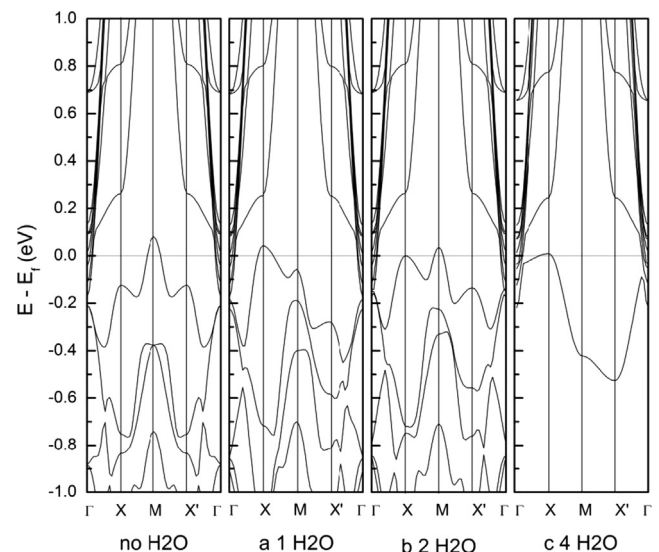


Fig. 2. Band structures of different models, from left to right: model without H_2O molecules, model a (1 H_2O), model b (2 H_2O) and model c (4 H_2O).

closer to the O atoms of the AlO_2 surface. Surface states can be affected by even one additional H_2O molecule per (1×1) surface unit. Valence bands in model c of Fig. 2 do not correspond to any bands in the film without water because those bands in model c are contributed by states of H_2O .

Fig. 3 shows the projected density of states (PDOS) of Ti with and without water molecules adsorbed on the LAO surface. Similar to Refs. 28, 29 the result shows that most of the conduction electrons occupy Ti d_{xy} orbitals since d_{xy} is lower in energy according to literature. In Fig. 3 no observable change from the PDOS of Ti can be found after adding water in models a and b. It is also illustrated that Ti d_{yz} and d_{xz} states overlap. The dipole field from H_2O would be screened by the dielectric LAO and it could not affect the interface directly. In Fig. 3, the Ti d_{xy} state in model c shifts down in energy at the interface, and it shows that the additional conduction electrons occupy mostly the Ti d_{xy} states at the interface.

Fig. 4 shows the PDOS of the water molecule and the AlO_2 group at the surface in models a and c, while the result of models b is similar to that of model a. As shown in Fig. 4(a), the peaks corresponding to the p_y (p_x) orbitals of the O_x (O_y) atom are right at the Fermi level, indicating the presence of holes on the surface without H_2O . It is interesting to see that the p_z orbital of O_y (O_x) in models a and b shifts upward in energy and beyond the Fermi-level. This should reflect the change of the surface hole states of models a–b in the band diagram as shown in Fig. 2. However, all O 2p states in model c are below the

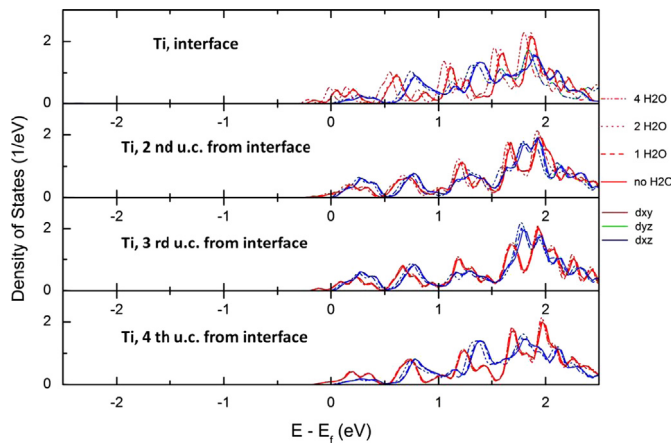


Fig. 3. (Color online) Projected density of states of Ti without or with water molecules on LAO surface (models a–c). PDOS of Ti d_{yz} and d_{xz} overlap.

Fermi level in Fig. 4(b), but a higher peak of the DOS around the Fermi level appears in the upper H_2O . This indicates that polarity discontinuity at LAO/STO interface is compensated by electrons transferred from H_2O , instead of the surface of LAO.

Fig. 5 is an estimation of the amounts of conduction electrons and holes determined by an integration of their corresponding PDOS around the Fermi level showing the result consistent with that of literature [30,31] where most conduction electrons are located at the interface, while away from the interface, conduction electrons decrease in amount. It is apparent that the density of conduction electrons n_{2d} is approximately equal to the density of holes n_h with the difference (error) less than 12.5% in each models. n_{2d} and n_h change less than the error in models a and b. Significant change happens in model c, where n_{2d} and n_h increase significantly by 52% and 32%, respectively. This result qualitatively matches the experimental work [17] where the density of conduction electrons n_{2d} increases from $1 \times 10^{13} \text{ cm}^{-2}$ to $2.5 \times 10^{13} \text{ cm}^{-2}$.

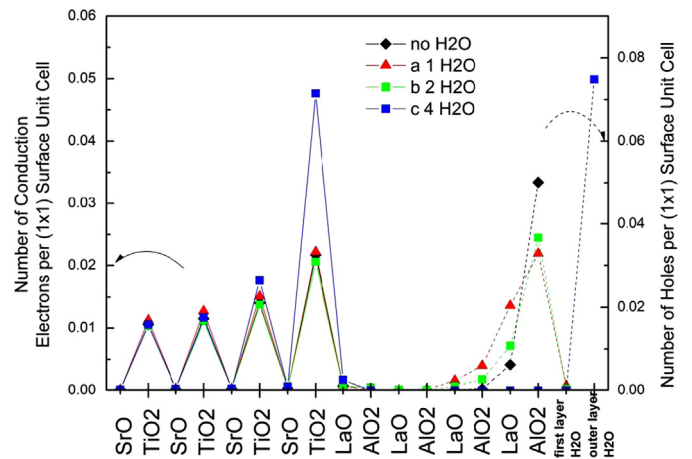


Fig. 5. (Color online) Distribution of conduction electrons with solid lines and holes with dashed lines.

Table 1
Dipole moment in different models.

Model	a	b	c
Change of dipole moment along $-z$ (eÅ)	−0.05	−0.08	0.23
Net H_2O dipole moment (eÅ)	0.05	0.05	0.59

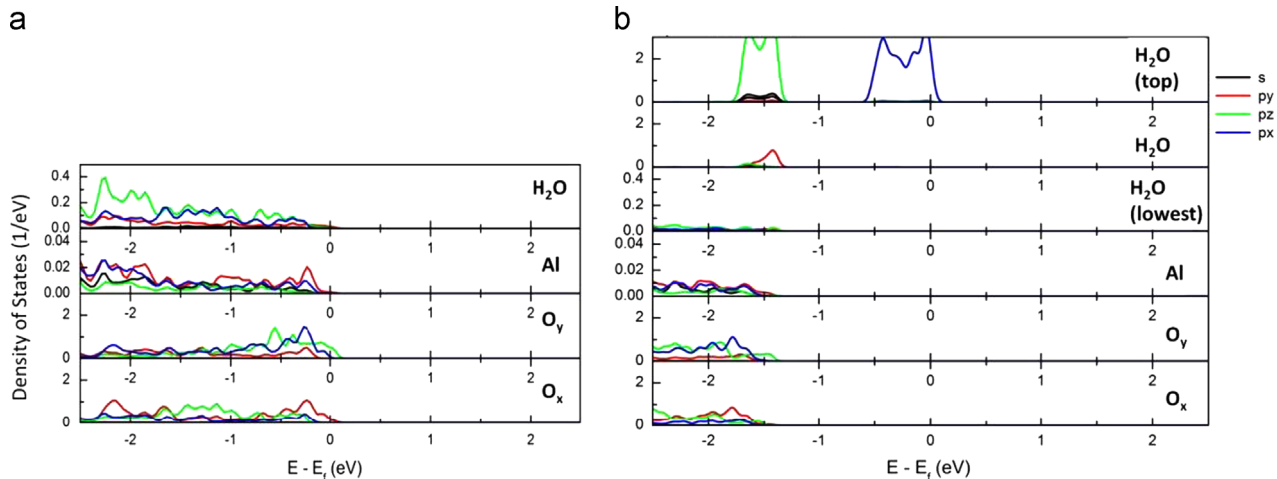


Fig. 4. (Color online) Projected density of states (PDOS) of H_2O and AlO_2 surface of models a (a) and c (b). The PDOS for model b are similar to (a).

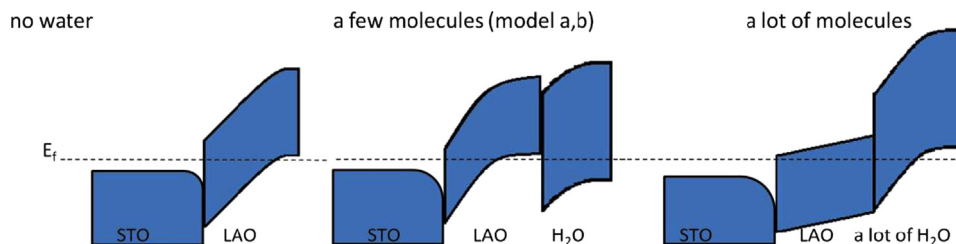


Fig. 6. (Color online) Sketching band structure along z for no water (left), a little water (middle) and lots of water (right).

The results illustrated in Fig. 5 can be intuitively interpreted by two points. First, the additional conduction electrons at the interface with H_2O should come mainly from the states (mostly O 2p orbitals) at the few topmost LAO atomic layers, since those states are sensitive to surface environment such as the different orientations of H_2O . Details of the dipole moment are listed in Table 1. Because the dipole field of the H_2O molecule is strong, the surface electronic structure should be affected as shown in Figs. 2–4. Secondly, after adding more H_2O molecules over the surface of LAO/STO, some H_2O molecules are at high electrical potential which results in a significant increase of 2DEG.

Fig. 6 shows the band structure diagram with water adsorbates outlined from PDOS. In the presence of a few H_2O molecules, the valence band rises in deeper layers of LAO as shown in the middle of Fig. 6(b), but it may be depressed on the surface. The energy levels of the H_2O molecules depend on their orientations, but the energy of the valence electrons of H_2O is not high enough to reach the Fermi-level, so 2DEG at the LAO/STO interface should be contributed only from LAO. From the results of calculation, densities of 2DEG and holes increase with the presence of more H_2O on the surface. When the electrical potential energy of H_2O molecules increases, the energy of valence electrons of H_2O may reach Fermi-level. Therefore, the valence electrons of the upper H_2O are of high energy and are transferred to the interface of LAO/STO, and thus increase of 2DEG can be observed [17].

4. Conclusion

Adsorption of H_2O molecules on AlO_2 -terminated surface of $\text{LaAlO}_3/\text{SrTiO}_3$ heterostructure has been investigated by first-principle studies. It reveals that adsorption of polar water molecules on the surface of LaAlO_3 can remarkably enhance the carrier density of two-dimensional electron gas at the interface. This change raises the possibility of various applications, such as polar molecule sensors.

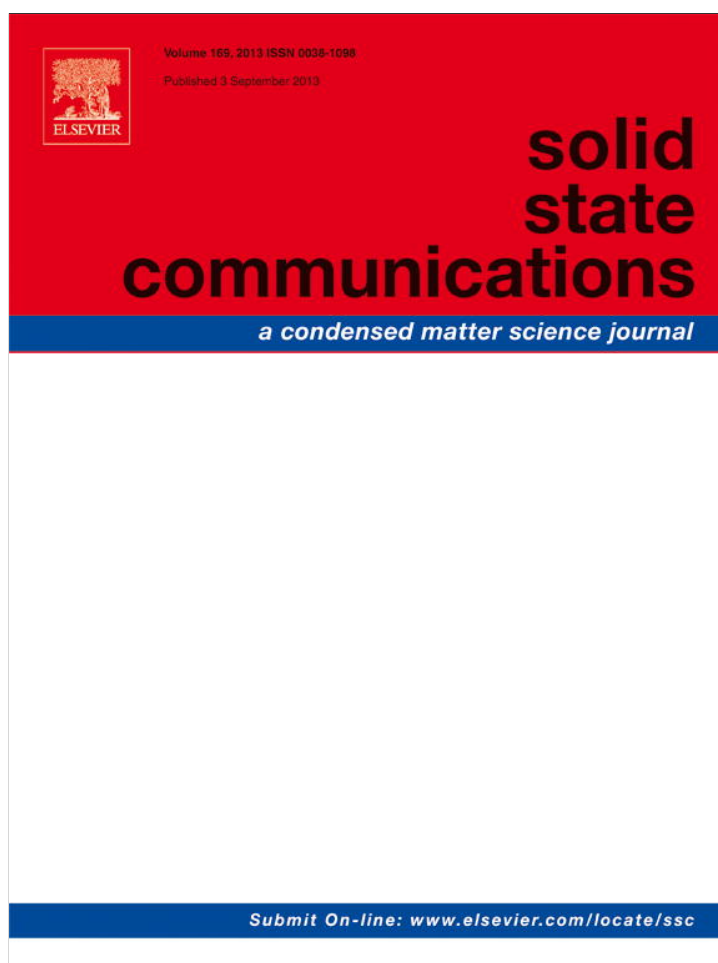
Acknowledgments

This research was supported by the National Key Basic Research Program of China under Grant no. 2013CB632900. This project is

also supported by Hong Kong GRF Grant (No. POLYU514512) and the Hong Kong Polytechnic University (No. G-YJ69).

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